

11. Vasoactive Molecules and the Kidney · assets 1-9 / 10

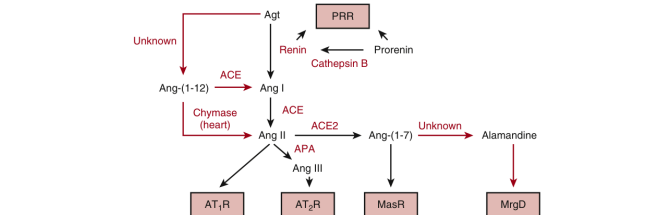


Fig. 11.1 Schematic depiction of the renin-angiotensin system components and selected actions. The enzymes of the system are shown in red. Newly described enzymatic pathways are shown as red arrows. Receptors are shown in the boxes. ACE, Angiotensin-converting enzyme; Agt, angiotensinogen; Ang, angiotensin; APA, aminopeptidase A; AT₁R, angiotensin type-1 receptor; AT₂R, angiotensin type-2 receptor; MasR, Mas receptor; MrgD, Mas-related G protein-coupled receptor; PRR, (pro)renin receptor. (Modified from Carey RM. Newly discovered components and actions of the renin-angiotensin system. *Hypertension*. 2013;62: 818-822.)

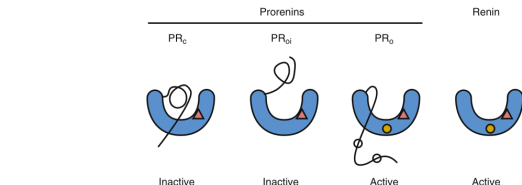


Fig. 11.2 The conformational changes and the expression of immunoreactive epitopes associated with the activation of prorenin are depicted. The main body of the molecule (*blue*), the substrate-binding cleft, and the prosegment (*black line*) are shown. The *closed triangle* represents the epitope of the main body expressed by PR₁ (prorenin in the inactive closed conformation), PR₂ (prorenin in the inactive intermediate conformation), PR₃ (prorenin in the open conformation), and renin. The *closed circle* (*yellow*) represents the epitope of the main body expressed by PR₁ (prorenin), but not by PR₂ (prorenin), PR₃ (prorenin), and renin. The *open circles* represent epitopes of the prosegment expressed by PR₁, but not by PR₂ and PR₃. (Modified from Schälchekamp MA, Derks HW, Deinum J, et al. Newly developed renin and prorenin assays and the clinical evaluation of renin inhibitors. *J Hypertension*. 2008;26:928-937.)

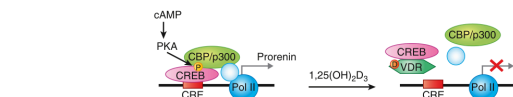


Fig. 11.3 Model of $1,25(\text{OH})_2\text{D}_3$ -induced transrepression of renin gene expression. The cAMP–PKA pathway activates CREB by phosphorylation, leading to recruitment of CBP/p300. In the presence of $1,25(\text{OH})_2\text{D}_3$, liganded VDR interacts with CREB and blocks its binding to CRE, leading to reduction of renin gene transcription. *cAMP*, Cyclic AMP; *CBP*, CREB-binding protein; *CRE*, cAMP response element; *CREB*, CRE-binding protein; *D*, $1,25(\text{OH})_2\text{D}_3$; *P*, phosphorylation; *PKA*, protein kinase A; *Pol II*, RNA polymerase II; *VDR*, vitamin D receptor. [Modified from Yuan W, Pan W, Kong J, et al. $1,25$ -Dihydroxyvitamin D_3 suppresses renin gene transcription by blocking the activity of the cAMP response element in the renin promoter. *J Biol Chem*. 2007;282:29821–29830.]

Fig_11_1

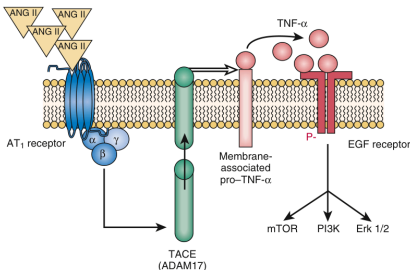


Fig. 11.4 Angiotensin II (**ANG II**) binds to its angiotensin II type 1 (**AT₁**) receptor, a G protein-coupled receptor lacking intrinsic tyrosine kinase activity. Through as-yet-undiscovered mechanisms, this interaction leads to the translocation of the metalloprotease tumor necrosis factor- α (**TNF- α**)-converting enzyme (**TACE**) from the cytosol to the cell surface, where it cleaves TNF- α from its membrane-associated pro-molecule, allowing it to bind and activate the epidermal growth factor (**EGF**) receptor. *Erk 1/2*, intracellular signal-regulated kinases 1 and 2; *mTOR*, mammalian target of rapamycin; *P13k*, phosphatidylinositol-3-kinase. (Modified from Woll G. "As time goes by": angiotensin II-mediated transactivation of the EGF receptor comes of age. *Nephrol Dial Transplant*. 2005;20:2050–2053.)

Fig_11_2

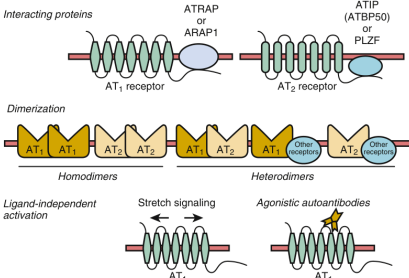


Fig. 11.5 Regulation of angiotensin receptors. *ARAP1*, AT₁ receptor-associated protein 1; *ATBP50*, AT₂ receptor-binding protein of 50 kDa; *ATIP*, angiotensin type 2 receptor-interacting protein; *AT₁ receptor*, angiotensin type 1 receptor; *AT₂ receptor*, angiotensin type 2 receptor; *ATRAP*, AT₁ receptor-associated protein; *PLZF*, promyelocytic leukemia zinc finger. (Modified from Mogi M, Iwai M, Horiuchi M. New insights into the regulation of angiotensin receptors. *Curr Opin Nephrol Hypertens.* 2009;18:138–143.)

Fig_11_3

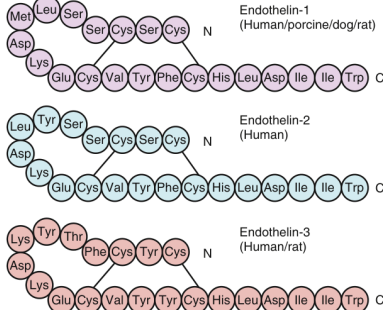


Fig. 11.6 Molecular structure of the three endothelin isoforms. (Modified from Schiffrin EL. Vascular endothelin in hypertension. *Vascular Pharmacol.* 2005;43:19–29.)

Fig_11_4

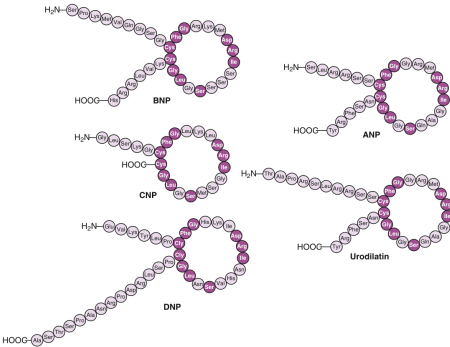


Fig. 11.7 Molecular structure of the natriuretic peptides. ANP, Atrial natriuretic peptide; BNP, brain natriuretic peptide; CNP, C-type natriuretic peptide; DNP, *Dendroaspis* natriuretic peptide. (Modified from Cea LB. Natriuretic peptide family: new aspects. *Curr Med Chem Cardiovasc Hematol Agents*. 2005;3:87–98.)

Fig_11_7

Fig_11_5

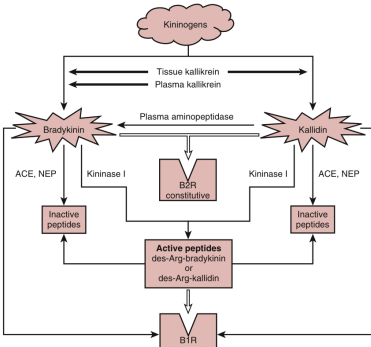


Fig. 11.8 Enzymatic cascade of the kallikrein-kinin system. *ACE*, Angiotensin-converting enzyme; *B1R*, bradykinin B1 receptor; *B2R*, bradykinin B2 receptor; *NEP*, neutral endopeptidase.

Fig_11_8

Fig_11_6

Stimulation	
Vasoeactive Peptides	Growth Factors
Angiotensin II	Epidermal growth factor
Bradykinin	Insulin-like growth factor
Vasopressin	Transforming growth factor- β
Endothelin-1	Coagulation
Epinephrine	Thromboxane A_2
Insulin	Tissue plasminogen activator
Glucocorticoids	Other
Prolactin	Calcium
Inflammatory Mediators	Hypoxia
Endotoxin	Shear stress
Interleukin-1	Phorbol esters
Tumor necrosis factor- α	Oxidized low-density lipoproteins
Interferon- γ	
Inhibition	
Atrial natriuretic peptide	Prostacyclin
Brain natriuretic peptide	Protein kinase A activators
Bradykinin	Nitric oxide
Heparin	Angiotensin-converting enzyme inhibitors

Table_11_1

Table 11.2 Peptides That Have Been Described as Substrates for Neprilysin		
Atrial natriuretic peptide	Cholecystokinin	Interleukin-1β
Brain natriuretic peptide	Corticotropin-releasing hormone	β-Lipotropin
C-type natriuretic peptide	Dynorphins	Luliberin
Endothelin-1	Endorphins	Luteinizing hormone-releasing hormone
Bradykinin and kallidin	Endothelin-2	α-Melanocyte-stimulating hormone
Substance P	Endothelin-3	Neurokinin A
Angiotensin I	Enkephalins	Neuropeptide Y
Angiotensin II	N-Formylmethionine-leucyl-phenylalanine	Neurotensin
Angiotensin 1-7	Fibroblast growth factor-2	Oxytocin
Adrenocorticotrophic hormone	Gastric-inhibitory peptide	Peptide YY
Adrenomedullin	Gastrin-releasing peptide	Secretin
Amyloid-β peptide	Glucagon	Somatostatin
Big endothelin-1	Gonadotropin-releasing hormone	Thymopentin
Bombesin-like peptides	Incretins	Vasoactive intestinal peptide
Calcitonin gene-related peptide	Insulin B chain	Vasopressin

Table_11_2